

OHDLL:Optimized Hybrid Deep Learning Model for Classification of Leukaemia images

Binju Saju

Department of Computer
Science - AI Engineering
Adi Shankara Institute of
Engineering & Technology
Kalady,Kerala
binjusaj@gmail.com

V Asha

Master of Computer
Applications, *New Horizon*
College of Engineering
Bengaluru, India
asha.gurudath@gmail.com

Neethu Tressa

Master of Computer
Applications, *New Horizon*
College of Engineering
Bengaluru, India
neethutressa89@gmail.com

Suhas R A

Master of Computer
Applications, *New Horizon*
College of Engineering
Bengaluru, India
suhaskashyap18@gmail.com

Shah Yash Jagesh

Master of Computer
Applications, *New Horizon*
College of Engineering
Bengaluru, India
shahyash490@gmail.com

Sandhya V

Master of Computer
Applications, *New Horizon*
College of Engineering
Bengaluru, India
sandhyasv9591@gmail.com

Abstract— A serious haematological cancer that causes mortality and morbidity in people of all ages is leukaemia. The haematologists examine bone marrow and blood to determine Leukaemia. Techniques for manual blood testing that have been in use for a long period are frequently slow and produce less reliable diagnoses. The study proposes a hybrid model to classify leukaemia cell images. Initially, images are pre-processed which includes Image resizing, Gray scale conversion and noise removal using a combination of Weiner filter and Median filter. Further, images are enhanced using Gaussian Mixture based histogram equalization. Features of the images are extracted using Dense Convolutional neural network. An optimized ResNet101 model with improved Aquila optimization is used for classification. Performance of the models is compared with other studies in literature. The proposed model has obtained an F1 score of 98.98%, an accuracy of 98.20%, sensitivity of 98.9%, specificity of 98.1%, and precision of 98%.

Keywords— Diabetic retinopathy, Resnet, IAO, Gaussian Mixture, DCNN

I. INTRODUCTION

In the bone marrow, leukaemia is a clonal expansion of hematopoietic stem cells. Acute lymphoblastic, acute myelogenous, chronic lymphocytic, and chronic myelogenous are the four broad subtypes that physicians are most likely to encounter. Acute lymphoblastic leukaemia occur in children than adults are but other variants are seen in adults. Genetic predispositions and environmental variables, including exposure to ionising radiation, are some of the risk factors. Fever, weariness, weight loss, bone pain, bruising, or bleeding are examples of non-specific symptoms. Leukocytosis and other abnormally raised or depressed cell are commonly found in a complete blood count[1].

The hematopoietic stem cell condition known as acute myeloid leukaemia (AML) is characterised by a stop in

hematopoiesis, which leads to the building of a clonal population of malignant cells or blasts. If left untreated, this malignant change in hematopoietic stem cells causes a loss of normal hematopoietic function, which usually results in mortality weeks to months after its clinical presentation[2-3].

A serious haematological cancer that causes mortality and morbidity in people of all ages is leukaemia. 350 000 new cases were reportedly reported worldwide in 2012. Several variables, such as limited access to healthcare and misclassification as a result of experienced staff, make leukaemia diagnosis difficult. Leukaemia was one of the potential diseases that ML may be used to treat. Several publications have examined the various methods for classifying and segmenting different blood cells, including WBC. With early work on blood and cervical smears, interest in computer-based diagnosis systems has been growing for the past 60 years. The approach to computer-based diagnostics has evolved with the introduction of ML algorithms [4].

A potentially disruptive technology in the area of medical imaging is artificial intelligence (AI)[5]. To weave design solutions that satisfy ethical and regulatory requirements and to create AI-based algorithms that improve outcomes, quality, and efficiency, details of principles and applications of radiomics, ANN, ML, and DL is a crucial foundation. Additionally, a more comprehensive view of applications, chances, and difficulties from a programmatic perspective aids in the ethical and long-term adoption of AI solutions.

In the area of digital imaging, including computer-aided diagnosis (CAD), radiomics, and image analysis, the usage of ML has been growing quickly. Recently, a branch of ML known as deep learning in the computer vision sector arose and quickly gained popularity[6].

Deep learning (DL) has emerged as a potent tool for a variety of imaging-related tasks, including classification, prediction,

detection, segmentation, diagnosis, interpretation, and reconstruction. DNN were first introduced in the computer vision field, but they have expanded to applications in medical imaging[7].

Recently, ML tools have matured enough to meet clinical objectives, and as a result, companies, research teams, and clinical teams are collaborating to build clinical AI solutions[8]. Physicians will be given more power and decision-making in clinical settings will be sped up thanks to the accelerated capability of DL in illness diagnosis. The use of contemporary medical technology and the digitization of healthcare have produced vast quantities of medical photographs in recent years. For medical applications and a knowledge of the underlying biological process, novel DL methods and computational models for effective data processing, analysis, and modelling of the obtained data are essential.

A unique kind of artificial neural network (ANN) called deep learning is part of ML and is inspired by the multilayered structure of human cognition[9]. As it relates to the analysis of large amounts of healthcare data, deep learning is currently receiving a lot of interest. Even though artificial neural networks (ANNs) were first developed in 1950, their ability to solve real-world problems was severely constrained by vanishing gradient and overfitting issues, which made it difficult to train deep architectures and, more importantly, lack of training data.

A skilled expert physician or doctor is essential for manual identification of haematological illnesses like acute leukaemia for the more precise early detection. However, current developments in AI methods like deep learning and machine learning enable the development of haematological problem detection algorithms that are more precise[10]. Although the effects of noise, blurring, weak edges, intensity inhomogeneity, and cell overlapping are greatly reduced in deep learning approaches, the extraction of more significant deep features and more effective classification are both carried out within a single neural network system to achieve more accurate disease detection. Therefore, more effective computer-aided detection (CAD) systems helps in the earlier and more accurate diagnosis of disease. It helps the doctor make an accurate disease diagnosis and treatment plan in order to save precious lives.

II. LITERATURE REVIEW

Karthick et al. [11] demonstrate the use of image processing to develop a quick and inexpensive computer-aided technique for categorising blast cells in leukaemia. A cell separation algorithm is then utilised to separate the overlapping cells after the Otsu's approach has first been used to separate the WBC from microscopic images of blood smears. Then, various traits are extracted from the nucleus, cytoplasm, and the complete cell. Consequently, the overall accuracy for identifying leukemia-affected cells is 90.32%. As a result, utilising the existing technique, the classification accuracy of leukemia-affected cells that can be distinguished from unaffected ones has shown tremendous potential.

Ahmad et al.'s improved pipeline for subtype categorization of WBCs [12] presents a wrapper feature selection technique based on a customised quantum-inspired evolutionary algorithm (QIEA), using deep neural networks and transfer learning for feature extraction. In terms of searching the search space, this quantum-inspired algorithm performs better than conventional evolutionary algorithms. Several baseline classifiers were used to categorise the QIEA-obtained reduced feature vector. The suggested methodology was tested using a publically available dataset containing 5000 images of five different subtypes of WBCs. The suggested method keeps classification accuracy at about 99% while shrinking the feature vector by 90%.

A fresh dataset of 500 peripheral blood smear images, including normal, Acute Myeloid Leukaemia, and Acute Lymphoblastic Leukaemia images, is introduced by Abhishek et al. [13]. The dataset contains around 1700 malignant blood samples. By including photographs from a publicly accessible dataset and creating a heterogeneous dataset, the dataset's size is expanded. Most important challenge of the proposed work is the automatic binary classification task, which makes use of the heterogeneous dataset. The suggested study uses cutting-edge algorithms to accomplish binary as well as three-class classification problems. The suggested approach produced a ResNet accuracy of 95% for binary classification.

Rezayi et al.'s [14] method of separating leukemic from healthy cells in microscopic images was applied to a dataset gathered from a CodaLab competition. Two well-known deep learning networks, the residual neural network (ResNet-50) and the VGG-16, were employed. We altered the learning parameters as well as the weights for these two networks because they had already been trained with the values we supplied.

Sampathila et al.'s clever deep learning system uses microscopic images of blood smears as its input data [15]. In order to identify leukemic cells from healthy blood cells, this technique use convolutional neural networks (CNNs). The tailored ALLNET model was trained and put to the test using the microscopic pictures made available as open-source data. The Nvidia Tesla P-100 GPU technique was used to train the model on Google Collaboratory. This precise classifier achieved a maximum accuracy of 95.54%, specificity of 95.81%, sensitivity of 95.91%, F1-score of 95.43%, and precision of 96%. The suggested method may be utilised in the pre-screening phase of complete blood count (CBC) and peripheral blood tests to identify leukaemia cells.

III. PROPOSED METHODOLOGY

A. Dataset

Accurate diagnosis of acute lymphoblastic leukaemia (ALL), a malignancy with a high prevalence, necessitates invasive, costly, and time-consuming diagnostic procedures. Peripheral blood smear (PBS) pictures used for ALL diagnosis are

crucial in the first cancer screening process to distinguish cancer cases from non-cancer cases. Due to the non-specific character of ALL signs and symptoms, misinterpretation is a common concern when these PBS images are analysed by laboratory users.

The bone marrow laboratory at Taleqani Hospital in Tehran, Iran, is where the photographs for this collection were created. A total of 3256 PBS pictures from 89 suspects were included in this collection. ALL patients whose blood samples were collected, processed, and stained by knowledgeable lab personnel. This dataset is separated into the benign and malignant classes. Hematogenes make up the first group, whereas the ALL group, which includes the three malignant lymphoblast subtypes Early Pre-B, Pre-B, and Pro-B ALL, makes up the latter. All of the photos were taken with a Zeiss camera at a 100x magnification in a microscope, and they were all saved as JPG files. The specific types and subtypes of these cells were identified by a professional using the flow cytometry instrument. We also offer segmented photos after colour thresholding-based segmentation in the HSV colour space.

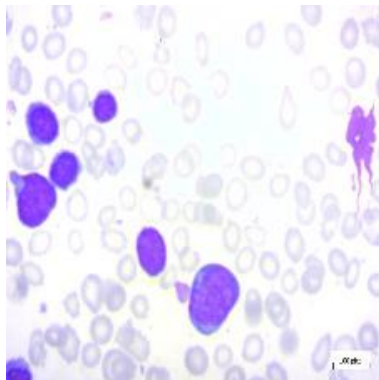


Figure 2. Sample image from dataset

B. Proposed Model

The proposed model is depicted in figure 3. Steps followed are:

- Pre-processing
 - Image resizing
 - Gray scale conversion
 - Noise removal- Wiener+ Median
 - Enhancement -GMCE
- Feature extraction using DCNN
- Model building - ResNet101_IAO

- Training and validation
- Model evaluation

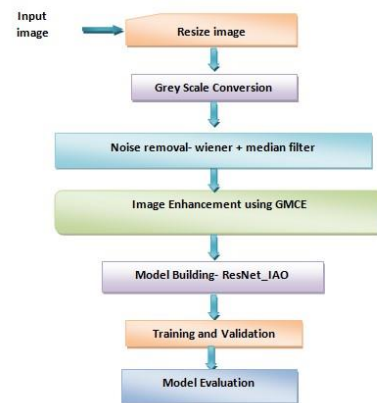


Figure 3. Proposed model

1) Image Pre-processing

Pre-processing is regarded as the most important component of every classification model since it raises the calibre of input, which raises the calibre of output.

Image resizing: Increasing the size of an image while maintaining its aspect ratio is known as image resizing, and it is a typical pre-processing step in machine learning. Resizing images helps introduce uniformity to the input data and improve training time. In many studies, downsizing the photos improved the model's accuracy. To make building the model considerably faster and simpler, all chest X-ray images in this work are shrunk to 224 224 pixels as follows:

$$I_{[224,224]}; J = \text{imresize} \quad (1)$$

This method takes the amount of pixels in the image that corresponds to the number of rows and columns and resizes the image to have 224 rows and columns.

Gray Scale Conversion: In image analysis and computer vision applications, converting an image to grayscale is a standard pre-processing step. It simplifies the algorithm and uses less processing power. It entails converting a colour image to a grayscale image where each grayscale pixel represents the brightness of its corresponding colour pixel. The following are the steps involved in grayscale conversion:

- 1) Scan the image to extract the RGB colour components into three 2D matrices;
- 2) create a zero matrix with the same order as the RGB image;
- 3) use the weighted sum method on the appropriate pixel position, such as (i,j), to convert each pixel value to a gray scale value as follow:

$$\text{grayscale value at } (i,j) = 0.299 \times R(i,j) + 0.587 \times G(i,j) + 0.114 \times B(i,j) \quad (2)$$

Noise Removal: Noise reduction in medical imaging is the primary function of image processing, which is an algorithmic

process that transforms an image. Noise reduces visual quality, which can compromise the precision of medical diagnoses.

There are a variety of filters that can remove the noise from medical photographs. However, due to its benefits and resilience over other non-linear filters, we picked the wiener filter and median filter combination in this investigation.

Median Filter: The non-linear filter category includes median filters. It is a non-linear filter with a lot of power. The variation between the adjacent pixels is lessened as a result. The replacement of the pixel value with the median value is the basic tenet of this. By placing the pixel values in ascending order and then computing the middle value, the median value is determined. For salt and pepper noise, use the median filter.

Wiener Filter : The wiener filter is used to take the noise out of the pictures. It operates on the idea of the best possible manner. The mean square error is computed. By comparing the received signal to the desired signal, noise reduction is achieved. The wiener filter's input is a constant. For the purpose of achieving the goal, it employs a statistical approach. Both the signal and the noise are taken to be linear stochastic processes. The wiener filter equation is provided below.

$$X_n \{u, v\} = W(u, v). Y(u, v) \quad (3)$$

Here $Y(u, v)$ is the received signal and $X_n(u, v)$ is the restored image. It uses for linearization of the blurring effect of the image. It helps in controlling the Signal to Noise ratio (SNR).

Image Enhancement: The wiener filter's input is a constant. It employs a statistical approach. Both the signal and the noise are taken to be linear stochastic processes. The wiener filter equation is provided below. GMCE is used to enhance the image after denoising. For simulating the pixels of the image, this model uses a Gaussian Mixture (GM) of the histogram. In a picture, every homogeneous region has a Gaussian shape structure. It is used to group shift and scaled Gaussians from the limited histogram of low contrast photos. The variance of these Gaussians is distributed throughout the partitions based on the average value, creating a global histogram. As a result, photos with low contrast have narrow histograms, and when the essential means are removed from one another, the contrast of the Both the number of areas and the visual quality of the image will rise.

Every arbitrary histogram is a weighted sum of I Gaussian that needs to determine 3 vectors of I parameters. They are scaling function $\mathcal{E}$, mean μ and variance σ^2 . Hence, the GM of histogram is given as:

$$H_{GM}^I(I | \sigma^2, \mathcal{E}, \mu) = \sum_{k=1}^I \mathcal{E}_k M(I | \sigma_k^2, \mu_k) \quad (4)$$

where $M(I | \sigma_k^2, \mu_k)$ is a k^{th} Gaussian probability density function (PDF) having the interval range of $I \in [0, P-1]$. Based on Equation (1), a height of every Gaussian is scaled using $\mathcal{E}_k$ and it is corresponding to place is filled by the respective region.

The variation between the GM and an original is optimized and it is expressed as:

$$\arg \min_{\sigma, \mathcal{E}, \mu} \sum_{I=0}^{P-1} H_o(I) - H_{GM}^I(I | \sigma^2, \mathcal{E}, \mu) \quad (5)$$

where H_o is an original image histogram.

For estimating the mean parameter, the GMCE models takes the intensity phase and it is expressed as:

$$\vec{\mu} = \arg \max_I \sum_{k=0}^{P-1} M(k | I, \sigma_0^2) H_o(k) \quad (6)$$

Then, for estimating the variance boundaries and sub-histogram are considered as inputs. After finding the sub-histogram, the scaling factor is estimated using the following Equation.

$$n_s^{t+1}(I) = n_s^t(I) - \vec{\mathcal{E}} M(I | \vec{\mu}, \sigma_0^2) \quad (7)$$

Finally, the intensity level is assigned as the output for histogram matching factor R for input S_1 . The enhancement model of GMCE is carried out completely by providing R for each pixel of low contrast images. Thus the image is enhanced efficiently by the GMCE model and used to classify diabetic retinopathy image.

2) Feature Extraction

The suggested approach involves feature extraction from the pre-processed photos to reduce the classification error. Dense CNN is preferred for extracting features for this reason. The Dense CNN model's last layer, which follows feature extraction, uses the given slit lamp pictures to classify the type of cataracts. The CNN approach produces superior outcomes in classification tasks in a number of recent studies. As a result, this method was employed in the proposed work to identify the cataract type.

The basic CNN design consists of five distinct layers: input, convolutional, max-pooling, fully connected, and output layers. Our suggested method also includes a number of dense layers that send to the activation function an evaluation of the weight average of each input. The dense layer will classify the image based on what the convolutional layers produce as their output. In Figure 4, the architecture is displayed.

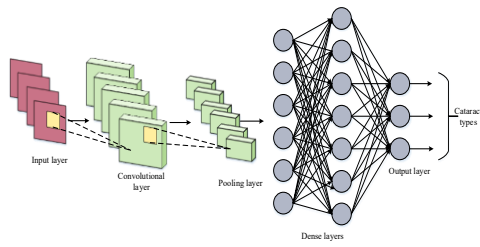


Figure 4 Structure of proposed DCNN method

The input layer of a CNN is the first layer in which images from a dataset are used as input. The second layer after this is the convolutional layer, which is created using different convolutional kernels. Utilising a back propagation method, the convolutional layer's parameters are optimised. The main goal of this layer is to extract as many features as possible from the slit lamp image while preserving complexity. The activation function is integrated and creates non-linear components in this convolutional layer. In the suggested neural network, the max-pooling layer is shown after the convolutional layer. This pooling layer creates feature mapping and lowers the dimensionality of the features. The collection of dense layers receives the data from the pooling layer. Each neuron in this layer is connected to every neuron in the layer before it. The activation function, which is referred to as a systematic layer, is primarily responsible for the functionality of CNN.

3) Model Building

The ResNet101_IAO model technique was employed in the proposed study to categorise cataract grades. In the proposed study, the ResNet101 and IAO algorithms are hybridised to produce better detection results. The basic ResNet is likewise a kind of CNN model, and its structure was influenced by the conventional VGG-19 approach. By disregarding the issues associated with collecting data from unbalanced instances, the suggested BE-ResNet101 recognises the grade. In the suggested work, the batch equivalence that has been described works in conjunction with ResNet101 to improve cataract grade identification outcomes.

The essentialResNet-101, a residual network with 101 layers, is an upgraded version of the ResNet-50 model. In the basic CNN design, various layers are interconnected, and the network is trained to perform a variety of tasks. The network extracts a wide range of attributes, and the output layer displays findings that are classified. ResNet's layers have the same amount of filters as CNN's convolutional layer and a similar size output feature amp. CNN's convolutional layer has 33 filters. The number of filters doubles if the feature

map's size is cut in half to reduce time complexity. By convolving layers with two strides, down sampling is accomplished directly in the ResNet model. The pooling layer, fully connected layer, and softmax layer are the final layers of this ResNet structure.

The residual learning can be described as minimization of input features obtained from that certain layer. This is performed by ResNet through shortcut connection to all the 33 filter pairs and is connected to the input of n^{th} layer to $(n + z)^{th}$ layer. By using activation from the preceding layer again until the layer nearest to the current layer has learned its own weights, distributive layers aim to prevent the problems associated with vanishing gradients. The weights will improve the upcoming layer while keeping the preceding layer inactive during the network training process. Thus, it is noted that the proposed ResNet model is effortless to training the samples than the traditional methods.

There are a total of 104 convolutional layers in the ResNet-101. It has 33 layers in all, and 29 of those layers use the output from the block before them. It is shown as residual connections in the diagram above, and these residuals are used as the first operand of the addition operator, which is used at the bottom of every block to get the input for the following blocks. The rested 4 blocks acquires the output of preceding block and utilize it in a convolutional layer with 1×1 of filter size and one stride subsequently by batch normalization layer. The normalisation procedure is made possible by this layer, and the finished product is sent to the addition operator at the block's output.

The available loss function affects the CRNN's detection performance. IAO is introduced to minimise the loss function. This strategy is population-based and was inspired by the way that Aquila hunts.

IAO has four stages for doing this:-

Expanded exploration- chose optimal features

Narrowed exploration- The search agent effectively targets the location of features. Then, it selects the crucial features and encircles them for the detection procedure.

Weight strategy- Each feature is located in a separate location inside the first search space, which is very extensive. Based on the maximum iteration, the distribution scope of the feature is compressed, creating a compressed search space. The suggested optimization strategy enters the local optimum since the feature set is less extensive. The qualities of the

various attributes are enhanced by the weight factor, which also aids in recovering the optimization procedure from the local optima.

Normalization- The method involves converting the dataset's numeric variable's value to a standard scale without distorting contrasts in the range of values. Batch processing in machine learning has the main advantage of increasing the stability and accuracy of model training and optimisation. Brightened input at each layer will aid in maintaining stable input distributions that will mitigate the negative effects of the internal covariate shift.

4) Training and Validation

The pre-processed and feature-extracted data is then used to train the chosen model. Training data and testing data were separated from the dataset for this investigation. 20% of the data images were utilised to test the system's performance, while the remaining 80% were used for training.

5) Model Evaluation

After training, a test dataset is used to evaluate the model's efficacy, and its performance is evaluated using a range of evaluation metrics. To ensure that the model can successfully generalise to novel, untested data, this is crucial. More information about performance evaluation is provided in the following section. In general, machine learning can be an effective method for detecting diabetic retinopathy conditions, but precise and reliable findings require careful data collection, pre-processing, feature extraction, and model selection, as well as stringent testing and validation.

IV. RESULTS AND DISCUSSION

Performance evaluation like are used to assess the suggested model. Table1 displays the obtained values. The proposed model has an F1 score of 98.98%, an accuracy of 98.20%, sensitivity of 98.9%, specificity of 98.1%, and precision of 98%.

Table 1. Performance evaluation of proposed model

Accuracy	98.20%
Sensitivity	98.90%
Specificity	98.10%
Precision	98%

F1 Score	98.98%
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The different models were found in the literature, and 5 of them were found to have the highest accuracy. These models were compared to the suggested mode based on the algorithms employed and the accuracy found. Table 2 and Figure 5 indicate the contrast.

Models	Model used	Accuracy
Karthick et.al [11]	otsu	90.32%
Ahmed et.al[12]	QIEA	95%
Abhishek et.al [13]	ResNet	95%
Rezayi et.al [14]	CNN	82.1%
Sampathila et. al [15]	Allnet	95%
Proposed Model	ResNet101 with IAO algorithm	98.20%

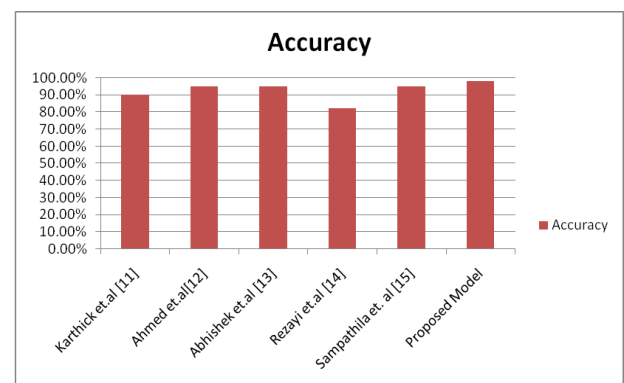


Figure 5. Performance comparison

V. CONCLUSIONS

Leukaemia is a severe haematological cancer that affects persons of all ages and results in mortality and morbidity. To identify leukaemia, haematologists check blood and bone marrow. Long-established manual blood testing procedures are frequently slow and result in less accurate diagnosis. A hybrid approach is suggested by the study to categorise leukaemia cell pictures. photos are first pre-processed, which includes shrinking the photos, converting them to grayscale, and removing noise using a mix of the Weiner and Median filters. Additionally, photos are improved utilizing histogram equalization based on a Gaussian Mixture. With the aid of a dense convolutional neural network, picture features are retrieved. For classification, an upgraded Aquila optimization

ResNet101 model is utilized. Performance of the models is compared with other studies in literature. The proposed model has obtained an F1 score of 98.98%, an accuracy of 98.20%, sensitivity of 98.9%, specificity of 98.1%, and precision of 98%.

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